Hyperspectral Image Classification via Multiscale Multiangle Attention Network

Jianghong Hu, *Graduate Student Member, IEEE*, Bing Tu, *Senior Member, IEEE*, Qi Ren, *Student Member, IEEE*, Xiaolong Liao, *Graduate Student Member, IEEE*, Zhaolou Cao, *Member, IEEE*, and Antonio Plaza, *Fellow, IEEE*

***Abstract*— Hyperspectral images (HSIs) provide a large amount of spatial and spectral information to characterize ground objects. However, they also contain a lot of redundant information, which makes it difficult to extract complex local and global spatial–spectral features. Considering that HSIs present multiscale similarity and anisotropic image features, multiscale and multiangle information can be used to effectively model local and global features and reduce the complexity of self-attention. This article proposes a new multiscale multiangle attention network (MMAN) for HSI classification that models the internal relationship between image features at local and global scales. First, three spectral–spatial feature extraction modules (at different scales) are constructed to extract the low-level features of the image. These modules are first used by a 3-D convolutional layer for spectral feature extraction and then input to a 2-D convolutional layer for spatial feature extraction. Next, the serialized tokens are input to the multiangle attention module. Finally, the learnable labels are identified through a linear layer, and the features of different scales are fused through a fully connected layer to realize the classification of samples. Experimental results on four standard datasets show that the proposed exhibits comparable or superior classification performance than other state-of-the-art methods.**

***Index Terms*— Anisotropy, cross-scale similarity, hyperspectral images (HSIs), multiangle attention, multiscale, transformer.**

NOMENCLATURE

X	Original HSI.
X_p	HSI after dimensionality.
P	Different scales of 3-D adjacent patches.
L	Number of tokens.
F	Features fused at multiple scales.
O	Ground object categories.
p	Size of each patch.
b_s	Batch size.
k_0	Number of 3-D convolution kernels.
k'	Number of 2-D convolution kernels.
Y_r	Final classification probability.
S_{HSI}	Output sequence of the HSI.
SA	Self-attention.
CSA	Correlation self-attention.

Manuscript received 16 November 2023; revised 29 January 2024; accepted 24 February 2024. Date of publication 28 February 2024; date of current version 13 March 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62271200 and in part by the Startup Foundation for Introducing Talent of Nanjing University of Information Science and Technology (NUIST) under Grant 2023r091. *(Corresponding author: Bing Tu.)*

Jianghong Hu and Xiaolong Liao are with the School of Information Science and Engineering, Hunan Institute of Science and Technology, Yueyang 414000, China (e-mail: hjhong911@163.com; xiaolongliao.china@gmail.com).

Bing Tu and Zhaolou Cao are with the Institute of Optics and Electronics, the Jiangsu Key Laboratory for Optoelectronic Detection of Atmosphere and Ocean, and the Jiangsu International Joint Laboratory on Meteorological Photonics and Optoelectronic Detection, Nanjing University of Information Science and Technology, Nanjing, Jiangsu 210044, China (e-mail: tubing@nuist.edu.cn; zhaolou.cao@gmail.com).

Qi Ren is with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen 518000, China (e-mail: ren_qi1113@foxmail.com).

Antonio Plaza is with the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, Escuela Politecnica, University of Extremadura, 10003 Cáceres, Spain (e-mail: aplaza@unex.es).

Digital Object Identifier 10.1109/TGRS.2024.3370919

I. INTRODUCTION

HYPERSPECTRAL images (HSIs) record continuous spectral information in almost contiguous wavelengths. Compared with traditional color or multispectral images, HSIs can provide a more detailed and comprehensive description of the scene. HSIs comprise tens to hundreds of bands in the visible and near-infrared spectral range. Each pixel in the HSI provides a spectral signature reflecting different physical and chemical properties of the observed object or ground features. HSI classification has a wide range of applications in agriculture, land use/land cover, urban environmental monitoring, mineral resource exploration, environmental ecological research, and so on.

HSI classification aims at accurately assigning pixels in the image to different ground object classes by analyzing the spectral and spatial information in the data. Traditional HSI classification methods use statistical and machine-learning techniques. Maximum likelihood classification [1], [2] is a commonly used method that assumes that the spectral distribution of each class of pixels follows a specific probability distribution and classifies the pixels by maximizing the probability of belonging to each class. Support vector machine (SVM) is another commonly used method [3], [4], [5], which constructs an optimal hyperplane for classification and maps the spectral information into a high-dimensional space to find the optimal classification boundary. Random forests (RFs) is an ensemble learning method that builds multiple decision trees and combines the results to achieve classification [6], [7]. However, these methods do not fully consider the spatial information in the HSI, resulting in classification results that

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.

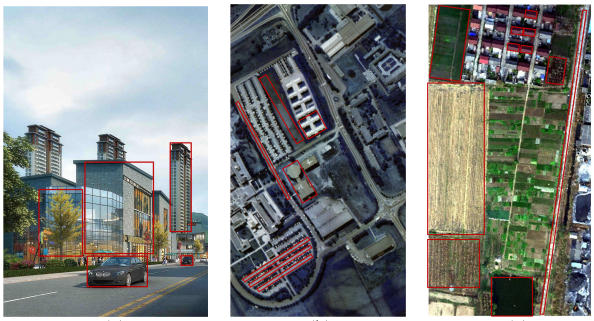
are affected by spectral mixing and spatial discontinuities. The combination of spatial and spectral information led to object-based classification methods. These methods first aggregate pixels into objects with some spatial continuity and then perform classification based on object-level features. Methods for spectral feature extraction and selection include principal component analysis (PCA) [8], [9] and linear discriminant analysis (LDA) [10], [11], [12]. Methods for spatial information fusion include pixel-based spatial filtering, object-based segmentation and feature extraction, and context-based classification methods. These methods can better capture the spatial–contextual information of ground objects and improve the accuracy and stability of classification. However, they suffer from high data dimensionality, unbalanced samples, spectral mixing, and spatial discontinuities.

In recent years, deep-learning technology has made significant progress in various fields. In remote sensing, many deep models have been proposed for HSI classification, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs) [13], deep neural networks (DNNs) [14], generative adversarial networks (GANs) [15], [16], and graph convolutional networks (GCNs) [17]. Among them, CNNs have become the preferred model for HSI classification due to properties such as multiscale feature extraction, local receptive field, and parameter sharing.

CNNs have been widely used in HSI classification due to their powerful feature extraction and classification capabilities. The work in [18] first proposed a deep CNN with five layers of weights to extract spectral features from HSIs in the spectral domain to perform classification. Subsequently, more CNN-based variants have been used for HSI classification tasks, such as the work in [19] that proposed a 2-D-CNN model to extract spectral and spatial features. In [20], [21], the 3-D-CNN was proposed for effective feature extraction. The work in [22] established that the association between different dimensions can be comprehensively modeled by fusing 1-D-CNN and 2-D-CNN, which increases the robustness of the model and effectively reduces the sensitivity to abnormal data. To make better use of the spatiotemporal information of HSIs, the work in [23] proposed a hybrid spectral CNN that combines 2-D-CNN and 3-D-CNN to effectively extract rich and comprehensive spatio-spectral feature information from multiple angles and reduces the complexity of the model. The work in [24] proposed an end-to-end spectral–spatial residual network (SSRN) for hyperspectral classification tasks, which can effectively alleviate the decline in the accuracy of the model and facilitate gradient backpropagation. The work in [25] proposed the differentiated-scale restricted GCN network, focusing on subtle adjacency relations between small regions. The work in [26], [27] proposed an HSI sharping method based on zero-shot learning and model-guided network (SSRNet) based on cross fusion. From the perspective of imaging sensor estimation and spectral super-resolution imaging, respectively, it is combined with CNN to fully extract the spectral–spatial features of HSIs. Paoletti et al. [28] introduced a deep pyramid residual network for HSI classification which gradually increased the diversity of high-level spectral–spatial attributes by increasing the dimension of the feature map, and effectively extracting the feature information found by the convolutional filter. The work in [29] proposed a residual spectral–spatial attention network to learn features for classification from useless bands. Zhao et al. [30] proposed the superpixel guided deformable convolution network, which performs convolution operation on the superpixel segmented image to fully extract the spatial and spectral features around the central pixel. Although CNNs (and their variants) have demonstrated good classification performance, they still suffer from shortcomings, such as the inability to model the temporal correlation of HSIs with time-series features. At the same time, the convolution kernel is limited by the local receptive field, which cannot fully capture the global spectral correlation and the spatial correlation.

To make the network pay more attention to global features and temporal correlation, the transformer model has been introduced into the research of HSI classification [31]. With the exploration of the application of the transformer model in HSI classification, researchers have begun to propose some variant models to adapt to the needs of this task [32], [33]. For example, He et al. [34] proposed the bidirectional encoder representation from transformer network (HSI-BERT) for HSI classification, which overcomes the influence of spatial distance to capture the global dependence between pixels. The context-aware representation is encoded by multihead self-attention, and the captured features fully capture the spatial and spectral information of the data. The sparse transformer was proposed by [35], which reduces the computational complexity and improves the efficiency of the model by introducing a sparse attention mechanism. The vision transformer (ViT) [36] regards the HSI as a 2-D feature, transforms the image pixels into a sequence, and inputs them into the transformer model for processing, capturing the relationship between pixels through the self-attention mechanism. The work in [37] proposed an attention-based second-order pooling network (A-SPN), which considers feature extraction from the perspective of second-order statistical features and adjusts complex hyperparameters. The work in [38] obtains spatial attention weights by measuring the correlation of pixel pairs and dynamically adapts to the changes of spatial pixels. The SpectralFormer (SF) [39] learns spectral local sequence information from neighboring bands, introduces cross-layer skip connections, and adaptively fuses the "soft" residuals across layers to preserve valuable information during layer-by-layer propagation. The work in [40] introduces a swin transformer that improves the performance of the transformer in processing large-size images through a hierarchical window mechanism. The work in [41] proposed a spatial–spectral transformer (SST) that first uses a CNN to extract spatial information and then uses a DenseTransformer to capture spectral information in HSIs. Sun et al. [42] proposed a spectral–spatial feature token-transformer to capture spectral–spatial features and high-level semantic features (SSFTT), which reduced the computational cost. The work in [43] proposed a multiattention transformer and adaptive superpixel segmentation-based active learning network. It not only pays attention to the global

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 1. Single scale input cannot fully capture the image details and distinguish mixed pixels. (a) Natural image. (b) University of Pavia dataset. (b) XuZhou dataset.

context information, but also retains the edge details. The work in [44] proposed a network-adaptive method to capture local spatial semantics based on local semantic feature aggregation.

The attention mechanism introduced by the above method realizes the fusion of local and global information by modeling the global information over a long distance, which can more effectively capture the feature information in HSIs and improve the classification performance, but it still has some limitations. For example, SF and SSFTT perform self-attention on global information, and the self-attention mechanism inherent in transformer models escalates computational complexity, this becomes notably evident with large-scale HSIs, demanding substantial computational resources and time. Furthermore, the self-attention mechanisms reliance on input sequence scale poses a challenge. Inputs of varying scales could result in disparate attention weight distributions. Considering that HSIs share similarities with natural images, where different features possess distinct size distributions, a singular scale input might inadequately capture image intricacies and struggle to differentiate mixed pixels (as depicted in Fig. 1). Consequently, when dealing with large-scale HSIs, the computational intricacies of the transformer model magnify, posing a substantial challenge for efficient processing.

HSIs, as natural images [45], possess anisotropic features. Therefore, to make full use of the transformer to obtain the spectral–spatial feature information, and according to the characteristics of different features with anisotropy, this article proposes a multiscale multiangle attention network (MMAN) model for HSI classification. First, the model uses multiple-scale patches that are fed as inputs to a two-layer CNN for spatial–spectral feature extraction. Second, the serialized tokens are input to a multiangle attention module and window self-attention module for feature representation and learning. Finally, the learnable label recognition and weight learning of the fusion features are carried out through the linear layer and the fully connected layer, respectively, to realize the final classification.

The main contributions of this article can be summarized as follows.

1) We propose a multiscale CNN fusion strategy for spatial–spectral feature extraction from HSIs, where each fusion strategy is composed of a 3-D convolutional layer and a 2-D convolutional layer that convolute the image under different receptive fields, thus capturing the features of objects at different scales and improving the robustness of the model.
2) A multiangle attention module is proposed to approximate the attention graph between the query and the key in self-attention by using the correlation point as an intermediate medium. Then, using HSI anisotropy, multiangle self-attention is carried out in rectangles or squares for efficient dependency modeling at both local and global scales.
3) By combining the multiscale CNN network and the multiangle transformer structure, the spectral–spatial information of the HSI is fully utilized to reduce the computational complexity and improve classification accuracy. Experiments and comparisons on four real HSI datasets validate the powerful classification ability of the proposed network.

The rest of this article is organized as follows. Section II introduces the relevant content of the proposed MMAN method. In Section III, the classification performance of the proposed method on four real HSI datasets is analyzed, and the classification results are compared with several other state-of-the-art classification methods. Section IV concludes the article with some remarks and hints at plausible future research lines.

II. Methodology

In this section, we describe the specifics of the proposed MMAN framework in detail. As shown in Fig. 2, the proposed method mainly consists of three parts, which are: multiscale local feature (MLF) extraction, multiangle global feature (MGF) extraction, and the multiscale feature fusion (MFF). The method is summarized in Algorithm 1. We provide a nomenclature of symbols appearing in Section II to elaborate on what each symbol represents.

A. MLF Extraction

The original HSI is represented as $\mathbf{X} \in \mathbb{R}^{a\times b\times l}$, where a, b, and l represent the width, height, and number of bands. The dimensionality of the original HSI is often high, and there is redundant information between bands. To effectively reduce the computational complexity, remove redundant information, and improve the generalization ability of the algorithm, PCA is used to reduce the dimensionality of the original data. The HSI after dimensionality reduction is represented as $\boldsymbol{X_p} \in \mathbb{R}^{a\times b\times c}$. To prevent excessive dimension reduction and thus the loss of key information, c is set to 30 in this article.

Considering that different ground objects have different sizes and shapes, the selection of multiscale features is beneficial to capture spectral and spatial information. To make full use of $\boldsymbol{X_p}$, we consider different scales of 3-D adjacent patches $\boldsymbol{P} \in \mathbb{R}^{p_s\times p_s\times c}(s = 1, 2, 3)$, being the size of each patch $\boldsymbol{p_s} \times \boldsymbol{p_s}$.

The number of generated 3-D patches is determined by the spatial size $\boldsymbol{a} \times \boldsymbol{b}$ of $\boldsymbol{X_p}$, and each patch ranges from $m_i - (p_s - 1)/2$ to $m_i + (p_s - 1)/2$ in with and $n_j + (p_s - 1)/2$ to $n_j - (p_s - 1)/2$ in height, with c spectral bands. A proportion

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.

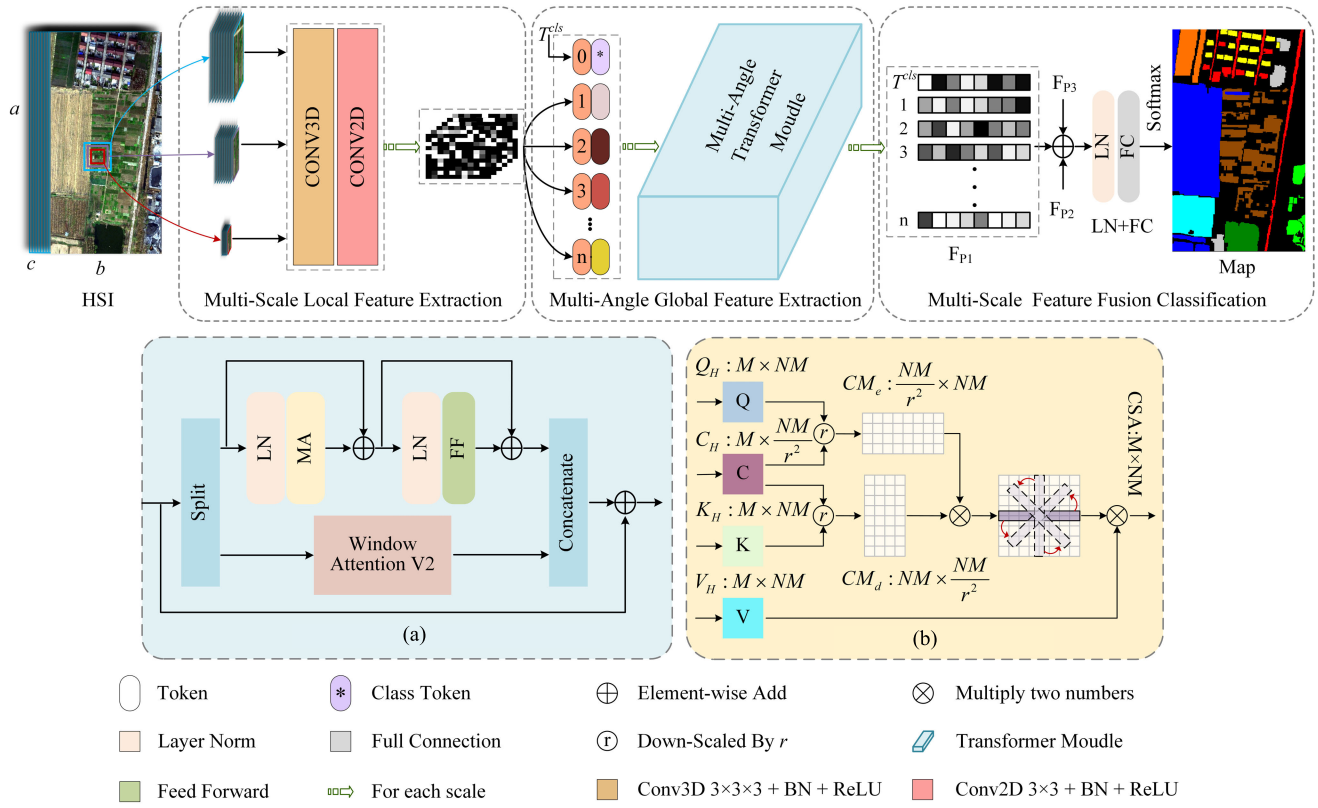


Fig. 2. Diagram of the proposed MMAN classification method. It comprises MLF extraction, MGF extraction, and MFF classification, where (a) multiangle transformer module and (b) multiangle attention.

$f\%$ of the samples are selected as inputs to the network, that is, the training set, and the remaining patches are used as the test set to verify the model's performance.

Next, 2-D and 3-D convolutional layers are used to extract the spectral–spatial information of the input patch. First of all, each training patch is fed to the 3-D convolution layer, where the value of the jth feature cube of the ith layer at the spatial location (α, φ, ρ) is denoted as $\upsilon_{i,j}^{\alpha,\varphi,\rho}$, with the expression

$$\upsilon_{i,j}^{\alpha,\varphi,\rho} = \phi\left(b_{i,j} + \sum_{z}\sum_{h=0}^{h_i-1}\sum_{w=0}^{w_i-1}\sum_{r=0}^{r_i-1}\omega_{i,j,z}^{h',w',r'}\upsilon_{i-1,z}^{\alpha+h',\varphi+w',\rho+r'}\right) \quad (1)$$

where $\phi()$ represents the activation function, $b_{i,j}$ denotes the bias, z denotes the number of feature cubes associated with the $(i-1)$ layer, and h_i, w_i, r_i denotes the width, height, and spectral dimensions of the convolution kernel, respectively. $\omega_{i,j,z}^{h',w',r'}$ denotes the weight parameter for the ith layer of the jth feature cube.

After the data has been processed in the 3-D convolutional layer, a rearrangement operation is performed to make it suitable for convolution in the 2-D convolutional layer. In the 2-D convolutional layer, the value of the jth feature cube of the ith layer at the spatial position (α, φ) is denoted as $\upsilon_{i,j}^{\alpha,\varphi}$, and the specific expression is

$$\upsilon_{i,j}^{\alpha,\varphi} = \phi\left(b_{i,j} + \sum_{z}\sum_{h=0}^{h_i-1}\sum_{w=0}^{w_i-1}\sum_{r=0}^{r_i-1}\omega_{i,j,z}^{h',w'}\upsilon_{i-1,z}^{\alpha+h',\varphi+w'}\right) \quad (2)$$

where h_i, w_i denote the width and height of the 2-D convolution kernel, and $\upsilon_{i,j}^{\alpha,\varphi}$ denotes the weight parameter of the jth feature map of the ith layer. The rest of the parameters have the same meaning as in (1).

In this model, the number of 3-D convolution kernels is k_0, the size is $k_w \times k_h \times k_c$, and the size of the feature cube after convolution is $(p_s - k_w + 1) \times (p_s - k_h + 1) \times (c - k_c + 1)$. 2-D convolution kernels are k', the size is $k_w \times k_h$, and the size of the feature map after convolution is $[p_s - 2(k_w + 1)] \times [p_s - 2(k_h + 1)]$. In this article, the 3-D and 2-D convolution kernels are set to $3 \times 3 \times 3$ and 3×3, respectively.

B. MGF Extraction

CNNs are good at extracting local features and hierarchical feature representations of an image, but are not flexible enough to handle global relationships and long-range dependencies. In contrast, the transformer model has a self-attention mechanism that models relationships between arbitrary locations, making it more suitable for capturing global information and long-range dependencies. Combining convolved features with a transformer allows for a better understanding of the relationships between different bands, especially in terms of spectroscopic correlations, by utilizing the transformer's relational modeling capabilities while retaining the ability to extract convolved features. Inputting the convolved features into the transformer requires processing, because the features produced by convolution are 2-D, whereas the transformer is designed to process sequential data. There are differences in data structure and processing between the two, so the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Multiscale Multiangle Attention Network

Input: Input the dimensionally reduced HSI data $\boldsymbol{X_p} \in \mathbb{R}^{a\times b\times c}$ and ground truth $\boldsymbol{X_t} \in \mathbb{R}^{a\times b}$, patches with size p_s of different scales, and the proportion of training samples $f\%$

1: Input the downscaled HSI data X_p, set up multiscale patches based on X_p, and divide the samples into training test sets according to the samples.
2: The batch size is set as $b_s = 64$, the number of epochs e is set to 120, and the learning rate is set to 0.001.
3: **for** $i = 1$ **to** e **do**
4: 3D-2D convolution operations are performed at different scales.
5: The 2D convolutional feature map is transformed into a 1D vector.
6: Concatenate multiple 1-D vectors and add position encoding to form the input sequence.
7: Execute multiangle transformer module.
8: LN and FC are used to fuse the output multiscale features.
9: Generate a categorical probability distribution using the softmax function.
10: The loss between the predicted output of the model and the actual label is calculated, and the weights are updated.
11: **end for**

Output: Save the trained model and put the test set into the model for classification prediction to get the predicted labels.

convolved features need to be converted into input format suitable for the transformer.

First, a flattening operation is performed to obtain a long 1-D vector by spreading the pixel values of each 2-D feature map produced by the convolution. In this way, the 1-D vectors of multiple feature maps can be concatenated into a longer sequence that serves as an input to the transformer. Considering that the transformer model lacks the positional information that is natural in a CNN, positional information needs to be introduced by adding positional coding. Position encoding is a set of vectors used to represent each position in the sequence that will be added to the input embedding. While adding the position information, the class markers T^{cls} need to be added to introduce an explicit start marker in the model, which can help the model understand the start position of the input sequence. The flow is as follows:

$$S_{\text{HSI}} = \left[T^{\text{cls}}, T^1, \ldots, T^L\right] + E_{\text{pos}} \tag{3}$$

where S_{HSI} denotes the output sequence of the HSI and L denotes the number of tokens. The embedded sequence is input into the transformer for semantic feature extraction and learning.

The self-attention mechanism is one of the core mechanisms of the transformer model for capturing dependencies between elements in sequence data. It allows each position in the sequence to generate a comprehensive representation by weighting information from other positions. Specifically,

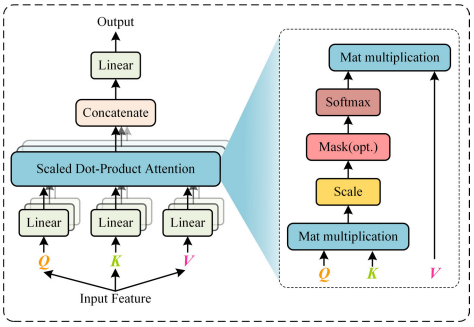


Fig. 3. Illustration of the multihead self-attention model.

we first obtain three representations: Query, Key, and Value (denoted by Q_H, K_H, V_H, respectively) by applying a linear transformation to each token of the input feature sequence S_{HSI}. The linear transformation is achieved by weight matrix multiplication. Then, the Query of each token and the Key of all tokens are dot-product, and the results are scaled using a normalization operation. After that, the attention weights of each token and other tokens are obtained through the Softmax function, which represents the degree of association between different tokens. Finally, these attention weights are multiplied with Value and weighted and summed to obtain the final self-attention representation, as shown in Fig. 3. This process can be understood as a weighted summary of the original sequence according to the importance of different tokens. The self-attention representation formula is as follows:

$$\text{SA} = \text{Attention}(Q_H, K_H, V_H) = \text{softmax}\left(\frac{Q_H {K_H}^T}{\sqrt{d}}\right) V_H \tag{4}$$

where $\text{SA} \in R^{L\times d}$ and d denotes the feature dimension. In addition, $AM = \text{softmax}((Q_H {K_H}^T/(d)^{1/2}))$ denotes the attention map.

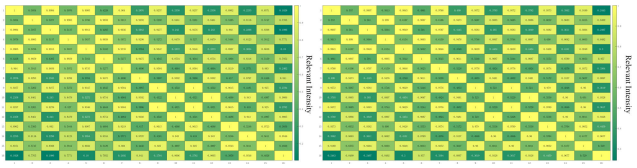
However, according to (4), it can be seen that the computational complexity of self-attention increases exponentially with the increase of the number of tokens, and it is easy to run out of memory when computing large-capacity images, so it is especially important to improve the efficiency of self-attention. Considering that HSIs possess cross-scale similarity property, that is, the similarity relationship between images at different scales, we introduce the concept of correlation point in the attention graph, which serves as a mediator for the comparison of the similarity between Q_H and K_H and can summarize the information of the image feature maps and possesses a low dimension. As shown in Fig. 4(a), it represents the correlation coefficient of the self-attention map of a feature block, and Fig. 4(b) shows the correlation coefficient of the attention map after adding the similarity points, from which it can be seen that the distributions of the two visualization maps are basically the same, and the Pearson correlation coefficient between the two is 0.9784, which also proves that there is a strong correlation between the two.

The correlation self-attention is modeled as follows:

$$\text{CSA} = V_H(\text{CM}_d \cdot \text{CM}_e) \tag{5}$$

$$\text{CM}_d = \text{softmax}\left(\frac{C_H \cdot {K_H}^T}{\sqrt{d}}\right) \tag{6}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 4. Example of an attention map, where (a) self-attention map and (b) similarity attention map. It can be seen that the two maps are very similar, indicating that the introduced similarity points reduce the computational cost without changing the original self-attention learning ability.

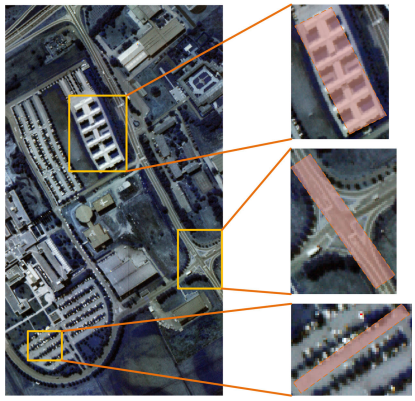


Fig. 5. Multiangle feature distribution on the University of Pavia dataset.

$$\mathrm{CM}_e = \mathrm{softmax}\left(\frac{Q_H \cdot {C_H}^T}{\sqrt{d}}\right) \tag{7}$$

where $M << L$, $C_H \in R^{M\times d}$ denotes the correlation point, $\mathrm{CM}_d \in R^{M\times L}$ denotes the attention graph between Q_H and C_H, and $\mathrm{CM}_e \in R^{L\times M}$ denotes the attention graph between C_H and K_H. Given that the number of correlation points is significantly less than the other tokens, it results in the size of the attention graphs CM_d and CM_e being much smaller than the attention graph. In (5), the self-attention is first computed for the correlation point and keys, and then proceed to the self-attention between the query and the correlation point, and finally the values are multiplied with the product of the two self-attention.

Meanwhile, the features of ground objects captured by HSI in different bands are often affected by anisotropy, that is, the reflection or radiation characteristics of ground objects may be different in different directions and angles. As shown in Fig. 5, in the University of Pavia dataset, the distribution of image features presents different angles. The distribution of most image features is not horizontal or vertical, but approximately presented as a 45° or 135° distribution. Therefore, it can be seen that multiangle attention in the global scope can effectively eliminate the redundant information caused by isotropic attention in the global scope according to the distribution of image features of different datasets and improve the accuracy and robustness of classification.

Therefore, we propose multiangle attention, which selects a region for special attention in the range of 0°–180°, and different angle attention is used alternately in the transformer to fully capture the image features, which not only ensures the modeling of image features from a global viewpoint, but also further reduces the computational complexity of self-attention. After combining correlated points and multiangle attention, we obtain correlated multiangle self-attention, as shown in Fig. 2(b). The correlation projection adopts average pooling plus linear projection. After projection, the resolution of image features in different directions is reduced by r times.

The multiangle transformer module layer is a key component of the HSI classification network framework, as shown in Fig. 1(a), which splits the extracted features equally along the channel dimensions by parallelizing the multiangle self-attention and window self-attention proposed in this article, and finally connects the features through the original channels. Here, the traditional self-attention mechanism is applied inside the window, and correlation computation is performed on the features inside the window, to fully capture the feature relationships inside the window and better understand the local details. Then, the feature maps of the two attention outputs are combined with the input features.

C. Multiscale Feature Fusion

MMAN consists of three modules (MLF, MGF, and MFF), which extract local features, global features, and MFF, respectively. MLF mainly uses the characteristics that the convolutional hybrid network is good at extracting local features and texture features of HSI, and learns the detailed features of the image. However, only using convolutional networks cannot focus on long-distance interspectral relationships, so MGF is used to capture the global context and long-range dependence of the image to better understand the connection between different bands. Then, the convolutional hybrid network is connected with the multiangle transformer end-to-end, so that the model can use local and global features at the same time, which helps to identify the target in the HSI more accurately. Finally, MFF is used to fuse and classify multiscale features. In the first part, the features of each scale processed by the Multiangle Transformer module are extracted, and then the features extracted at the three scales are added according to their dimensions to obtain the fused features. The first class token of the fused features is extracted, and layer Normalization is then used to normalize the class tokens to ensure that they have a similar distribution. The second part uses a fully connected layer to learn the weights of the fused features and maps each feature to a shared feature space with O ground object categories. The fusion process is expressed as follows:

$$F = F_{P1} \oplus F_{P2} \oplus F_{P3}. \tag{8}$$

The Softmax activation function is applied to the fused feature map representation to change the O-dimensional feature into a 0-1 probability Y. The number of samples in each batch is defined as R, and Y_r is the predicted class output probability of the rth target pixel in a batch. The final classification probability result Y_r is expressed as

$$Y_r = \underset{o\in[1,O]}{\arg\max}\, F_r(o), \quad \text{s.t.}, r \in [1, R]. \tag{9}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.

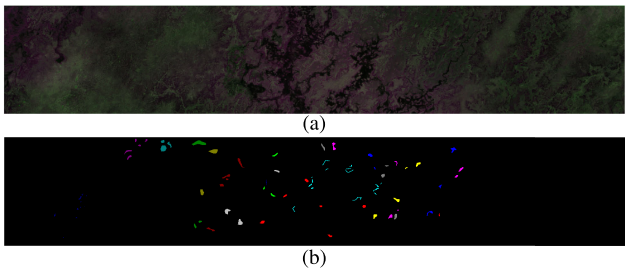


Fig. 6. Botswana dataset. (a) Composite image. (b) Ground truth.

III. Experimental Results and Analysis

All the classification experiments involved in this article were carried out on an Intel[1] Core[2] i9-10900KF CPU @ 3.70 GHz, NVIDIA GeForce RTX3070Ti GPU, and 32GB RAM. The MMAN model is implemented using the PyTorch library, and other comparison methods use their original experimental configuration environments.

In this section, we will conduct detailed experiments and analyses on the proposed MMAN to fully verify its performance and effectiveness.

A. Dataset Descriptions

This section introduces the four HSIs used in experiments (i.e., Pavia University (UP), Botswana, Houston 2013, and Xuzhou).

1) Botswana Dataset (BS): The Botswana dataset is an HSI dataset acquired by NASA's EO-1 satellite over the Okavango Delta in 2001. The dataset has a size of 1476 × 256 pixels, a spatial resolution of 30 m/pixel, and a total of 14 feature categories. After removing noisy bands, the remaining 145 bands were used. Fig. 6 shows a composite image and the ground truth. The names of land classes and the number of training and testing samples are shown in Table I.

2) Houston 2013 Dataset (HU): The Houston dataset was acquired by the ITRES CASI-1500 sensor and originally used in the IEEE GRSS Data Fusion Competition in 2013. The data size is 349 × 1905 pixels, the spatial resolution is 2.5 m/pixel, and there are 15 feature categories. It contains 144 bands with a spectral range of 0.364–1.046 μm. Fig. 7 shows the truth graph and false-color graph of the HU dataset. See Table II for detailed information on training and test samples.

3) UP Dataset: The UP dataset was collected in 2001 by the Reflective Optics Spectrographic Imaging System (ROSIS-03) over the city of Pavia, Italy [46]. The HSI has a size of 610 × 340 pixels, a spatial resolution of 1.3 m/pixel, and a total of nine different feature categories. It consists of 115 bands with a wavelength range of 0.43–0.86 μm. After removing 12 bands affected by noise, 103 bands are left for experiments. The UP dataset is shown in Fig. 8. A fixed number of samples were selected for training and testing, as shown in Table III.

[1]Registered trademark.
[2]Trademarked.

TABLE I
Number of Training and Test Samples of Each Class for the Botswana Dataset

Class No.	Class Name	Training	Test
1	Water	27	243
2	Hippo grass	10	92
3	Floodplain grasses1	25	226
4	Floodplain grasses2	22	193
5	Reeds1	27	242
6	Riparian	27	242
7	Fierscar2	26	233
8	Island interior	20	183
9	Acacia woodlands	32	282
10	Acacia shrublands	25	223
11	Acacia grassland	30	275
12	Short mopane	18	163
13	Mixed mopane	27	251
14	Exposed soils	9	86
Total Number		325	2923

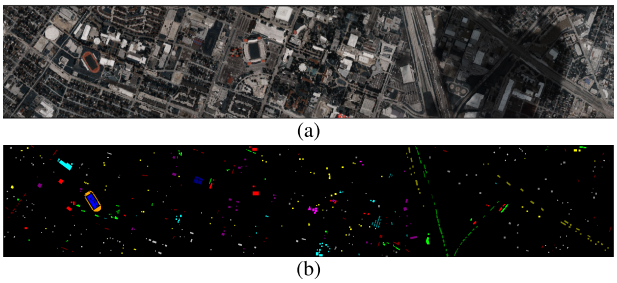


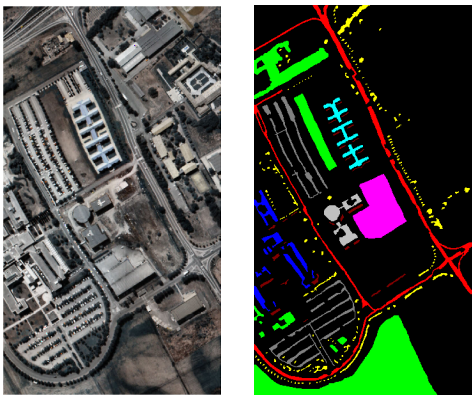
Fig. 7. Houston 2013 dataset. (a) Composite image. (b) Ground truth.

TABLE II
Number of Training and Test Samples of Each Class for the Houston 2013 Dataset

Class No.	Class Name	Training	Test
1	Healthy Grass	125	1126
2	Stressed Grass	125	1129
3	Synthetic Grass	70	627
4	Tree	124	1120
5	Soil	124	1118
6	Water	33	295
7	Residential	127	1141
8	Commercial	124	1120
9	Road	125	1127
10	Highway	123	1104
11	Railway	123	1046
12	Parking Lot 1	123	1112
13	Parking Lot 2	47	422
14	Tennis Court	43	385
15	Running Track	66	594
Total Number		1502	13166

4) XuZhou (XZ) Dataset: The Xuzhou dataset was collected by the HYSPEX hyperspectral airborne camera over the XuZhou suburban site in November 2014 [47], [48]. The dataset has a size of 500 × 260 pixels, a spatial resolution of 0.73 m/pixel, and a total of nine feature categories. After removing noise in the range of 0.4415–2.508 μm, 436 spectral bands were used (see Fig. 9). Specific training samples and test samples are shown in Table IV.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.

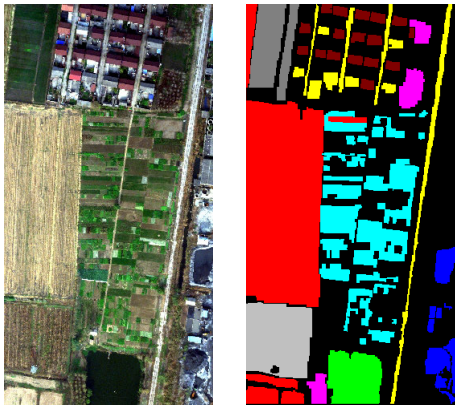


(a) (b)

Fig. 8. University of Pavia dataset. (a) Composite image. (b) Ground truth.

TABLE III
NUMBER OF TRAINING AND TEST SAMPLES OF EACH CLASS FOR THE UNIVERSITY OF PAVIA DATASET

Class No.	Class Name	Training	Test
1	Asphalt	332	6299
2	Meadows	932	17717
3	Gravel	105	1994
4	Trees	153	2911
5	Painted_m_s	67	1278
6	Bare_soil	251	4778
7	Bitumen	67	1263
8	Bricks	184	3498
9	Shadows	47	900
Total Number		2138	40638



(a) (b)

Fig. 9. XuZhou dataset. (a) Composite image. (b) Ground truth.

TABLE IV
NUMBER OF TRAINING AND TEST SAMPLES OF EACH CLASS FOR THE XUZHOU DATASET

Class No.	Class Name	Training	Test
1	Bareland-1	2640	23756
2	Bareland-2	403	3624
3	Crops-1	278	2505
4	Crops-2	521	4693
5	Coals	1318	11866
6	Concrete	244	2192
7	Trees	699	6291
8	Lake	478	4299
9	House-Roof	307	2763
Total Number		6888	61989

B. Experimental Setting

1) Implementation Details: To qualitatively and quantitatively compare the performance of the nine methods listed in this article on the four HSIs (i.e., Botswana, UP, Houston 2013, and Xuzhou), overall accuracy (OA), average accuracy (AA), accuracy per class, and Kappa coefficient were used to evaluate the performance of MMAN proposed in this article, where OA represents the total number of correct classifications in samples divided by the total number of samples and AA represents the number of correct predictions in each class divided by the total number of classes. The Kappa coefficient is used to measure the consistency test between the predicted results and the true labels. Each of the nine methods randomly selected the same proportion of training samples for ten repeated experiments to obtain the mean values of OA, AA, and Kappa. The number of training epochs is set to 120, and the best results are marked in each table by using bold typeface.

2) Comparison With State-of-the-Art Baseline Methods: The proposed method was compared with eight representative methods in recent years. All experiments were repeated ten times and the average value was taken, and the corresponding evaluation index value of each method was recorded. The comparison methods are the following ones.

1) *CDCNN [49]:* CDCNN is a deep CNN that combines neighboring pixel vectors to explore local context interaction, and its input patch size is set to $5 \times 5 \times B$, where B represents the dimension of the input spectrum.
2) *SSRN [24]:* SSRN is mainly composed of spectral and spatial residual blocks. It is a supervised deep-learning framework that can continuously learn discriminant features in HSIs. The network takes the original 3-D cube as input, and the input size is $7 \times 7 \times B$, where B represents the dimension of the input spectrum.
3) *FDSSC [50]:* FDSSC consists of two different dense blocks, which automatically extract spectral and spatial features. Its input size is $9 \times 9 \times B$.
4) *DBMA [51]:* DBMA is a densely connected double-branch multiattention mechanism network based on 3-D-CNN. The channel-wise attention enhances the spectral features and the spatial attention focuses on the region with the largest amount of information. The entered patch size is $9 \times 9 \times B$.
5) *DBDA [52]:* The DBDA structure is composed of two dense blocks (dense spatial block and dense spectral block). Finally, the classification results are obtained by the fusion of spectral feature and spatial feature mapping. The entered patch size is $9 \times 9 \times B$.
6) *SF [39]:* SF is a kind of transformer structure that can learn the local sequence information of the spectrum from the neighbor bands of the HSIs and design a cross-layer jump connection. The input patch size is $7 \times 7 \times 3$. Due to insufficient computing memory, the input patch size for the Xuzhou dataset in this article is $5 \times 5 \times 3$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.

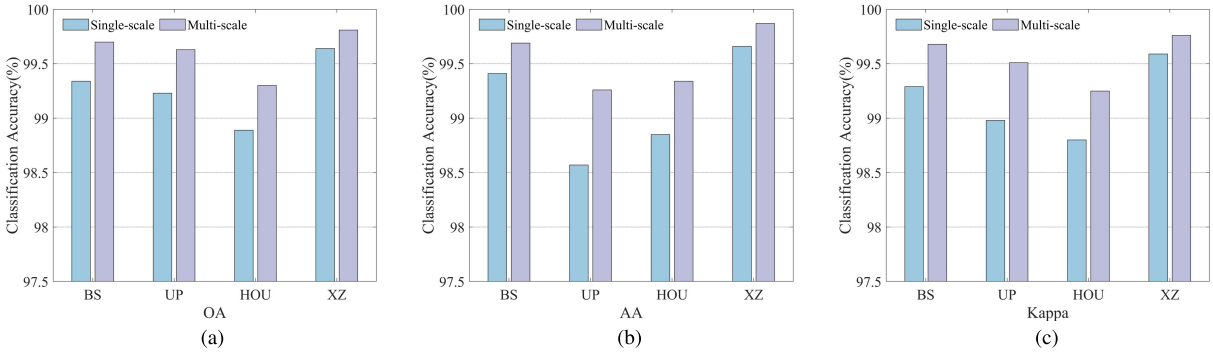


Fig. 10. Comparison of multiscale and single-scale classification performance. (a) OA. (b) AA. (c) Kappa.

TABLE V
EFFECT OF DIFFERENT PATCH SIZES ON THE OA PERFORMANCE OF THE MODEL ON THE FOUR DATASETS

Patch size	BS	UP	HU	XZ
5×5	96.18	95.66	93.55	97.63
7×7	98.06	98.38	97.93	98.93
9×9	99.12	98.71	98.72	99.43
11×11	**99.34**	**99.23**	98.77	98.59
13×13	98.56	99.1	**98.89**	**99.68**
15×15	98.41	99.02	98.53	99.68
17×17	98.43	98.57	98.19	99.64

7) *A-SPN [37]:* A-SPN is very powerful by using second-order statistics and data-adaptive attention weights to model discriminative and representative features. According to the setting of the original paper, we choose the patch size as 9 × 9.
8) *SSFTT [42]:* SSFTT method consists of a spectral–spatial feature extraction module, a Gaussian weighted feature tokenizer, and a transformer encoder module, which are mainly used to capture spectral–spatial features and high-level semantic features. The entered patch size is 13 × 13 × 30.
9) *SSMTr [53]:* SSMTr has two training strategies. In the first stage, the local perception bias is embedded into the pretrained transformer by reconstructing the input, and the fully connected layer and the trained Transformer cooperate to classify HSIs. The parameters adopted were set as in the original paper.

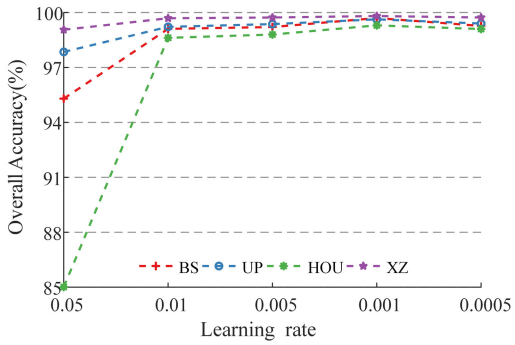


Fig. 11. Model performance at different learning rates.

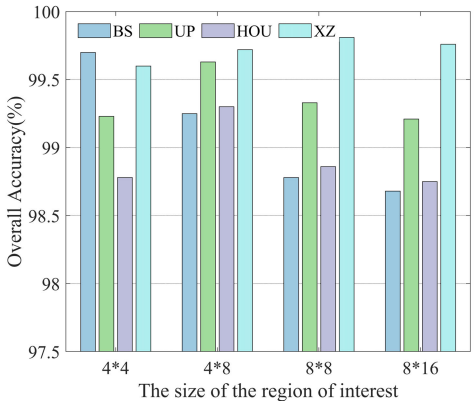


Fig. 12. Effect of ROI size on model classification performance.

C. Parameter Analysis

This section focuses on the analysis of the impact of three parameter variations, learning rate, patch size, and region of interest (ROI) size, on the performance of the MMAN method.

1) Effect of Patch Size: In HSI classification, the input patch size should match the target scale and context relationship in the image. If the target object has a large scale in the image, then selecting a larger patch helps to include the full picture and context information of the target. If the target object scale is small, smaller patches can be selected to improve the local details of the features. The four datasets contain different spatial resolutions of different sizes and different scene complexities, which are very suitable for analyzing the influence of Patch size on the classification performance of the model. The detailed results are shown in Table V. It can be seen that the performance of classification accuracy on the four datasets approximately increases first and then decreases when the input patch increases from small to large. The classification performance of UP and BS datasets is the best when the input is 11 × 11, and the classification performance of HU and XZ datasets is the best when the input is 13 × 13.

Although smaller patches can provide richer local information, they may lead to features that are not sufficient to capture the global structure of the whole image. However, larger patches can provide more global information, but the computational complexity may increase significantly [54]. Especially for images with complex feature distribution (such

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ACCURACY COMPARISON OF DIFFERENT METHODS ON THE BS DATASET

Class No.	Accuracy of classification obtained by different comparison methods (in %)									
	CDCNN	SSRN	FDSSC	DBMA	DBDA	SF	A-SPN	SSFTT	SS-MTr	MMAN
1	**100.00**	99.66	98.95	98.75	99.11	**100.00**	**100.00**	99.70	**100.00**	**100.00**
2	97.39	98.84	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	99.33	**100.00**	**100.00**
3	95.59	**100.00**	**100.00**	**100.00**	**100.00**	98.23	99.34	99.96	98.67	**100.00**
4	98.66	99.45	98.84	99.76	99.04	94.82	**100.00**	99.53	98.97	**100.00**
5	92.67	97.51	97.49	97.06	95.58	87.60	98.54	**98.55**	97.53	**98.55**
6	96.21	99.17	99.16	98.33	**100.00**	79.33	99.17	98.00	**100.00**	98.33
7	99.88	**100.00**	**100.00**	**100.00**	**100.00**	95.28	**100.00**	99.91	**100.00**	**100.00**
8	98.90	94.02	99.90	98.93	**100.00**	**100.00**	**100.00**	99.95	**100.00**	**100.00**
9	95.56	**100.00**	**100.00**	**100.00**	**100.00**	98.94	**100.00**	99.46	**100.00**	**100.00**
10	97.24	98.64	**100.00**	99.11	**100.00**	95.52	95.52	99.54	**100.00**	**100.00**
11	99.48	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	99.63	**100.00**	**100.00**
12	97.87	99.83	**100.00**	**100.00**	**100.00**	**100.00**	99.08	98.38	**100.00**	**100.00**
13	98.84	**100.00**	**100.00**	99.21	99.55	80.91	**100.00**	99.96	98.91	**100.00**
14	**100.00**	**100.00**	99.74	**100.00**	98.46	**100.00**	**100.00**	99.02	91.43	98.83
OA(%)	97.50	99.23	99.37	99.29	99.48	99.38	99.45	99.38	99.32	**99.70**
AA(%)	97.74	99.18	99.51	99.37	99.55	99.49	99.48	98.35	98.91	**99.69**
Kappa	97.29	99.17	99.32	99.23	99.43	99.33	99.41	98.30	99.26	**99.68**

as the HU dataset), selecting a larger patch cannot fully capture the detailed features, and selecting a smaller patch will lead to information loss. Therefore, using multiscale patch input can capture the features in the image more comprehensively, improve the diversity and information of the features, and help the model better understand the structure and correlation inside the image. Specifically, setting up the basic spatial dimension parameters τ, corresponding to the input of three data space size of $(2\tau+1)\times(2\tau+1)$, $(2\tau+3)\times(2\tau+3)$, $(2\tau+5)\times(2\tau+5)$, $\tau\in[2,5]$. For the UP and BS datasets, when the input $\tau=4$ for the classification of the best, for HU and XZ datasets, when the input $\tau=5$, better performance is obtained. The final single- and multiscale comparison results are shown in Fig. 10. From the results of the four datasets, it can be seen that the multiscale classification performance is more conducive to extracting the features of HSIs.

2) Effect of Learning Rate: Learning rate is a crucial hyperparameter in deep learning. For HSI classification problems, an appropriate learning rate setting plays a key role in the success of training and model performance. If the learning rate is set too large, the model may diverge, leading to training failure. On the other hand, if the learning rate is set too small, the model will train very slowly and may even get stuck in a local minimum and fail to converge to the desired solution. Therefore, the setting of the learning rate needs to be carefully analyzed and adjusted to ensure that the training process can be carried out effectively and produce a high-performance classification model. Considering the complexity of the model and the characteristics of the dataset, Fig. 11 shows the classification performance experiments of the proposed framework with different learning rates set based on four datasets. Experimental results show that, when the learning rate is set to 0.001, the classification performance of the model is the best of the four datasets.

3) Effect of the Size of the Region of Interest: When performing multiperspective self-attention, the selection of the size of the attention region is very important, which needs to make a tradeoff between task requirements, computational complexity, and performance. We experimentally explored

the performance of different sizes of ROIs on the classification task, as shown in Fig. 12. On the BS dataset, the OA performance of MMAN shows the best performance when a 4×4 size region of attention is selected. The possible reason is that most of the ground object distributions in the BS dataset are relatively small. On the UP and HU datasets, the OA performance of the model is the best when regions with the size of 4×8 are selected, which is because the distribution of ground objects in these two datasets is large or small, and the selection of too small or too large regions will have an impact on feature extraction. On the XZ dataset, the classification performance of the model is better than that of others when the size of regions is 8×8. This is because the distribution of ground objects in this dataset is dominated by large ground objects, and choosing a large region for attention is conducive to feature extraction and learning. When the size of 8×16 is selected for attention, the performance of the model on all datasets is decreased, which may be because the attention region is too large and easy to bring other redundant information, which affects the classification performance.

D. Comparison With Other Methods

To verify the classification performance of the MMAN model, we compare it with some representative methods in recent years and analyze it from a qualitative and quantitative perspective.

1) Results on the BS Dataset: On the BS dataset, we adopt 10% of the samples as the training set and the remaining 90% of the samples as the test set, the specific classification results are shown in Table VI, and the best-performing results are shown in bold. It can be seen that the overall performance of MMAN in three evaluation indicators OA, AA, and Kappa is better than the other eight comparison methods, and in the classification accuracy of various ground objects, such as Healthy Grass, Stressed Grass, and other

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.

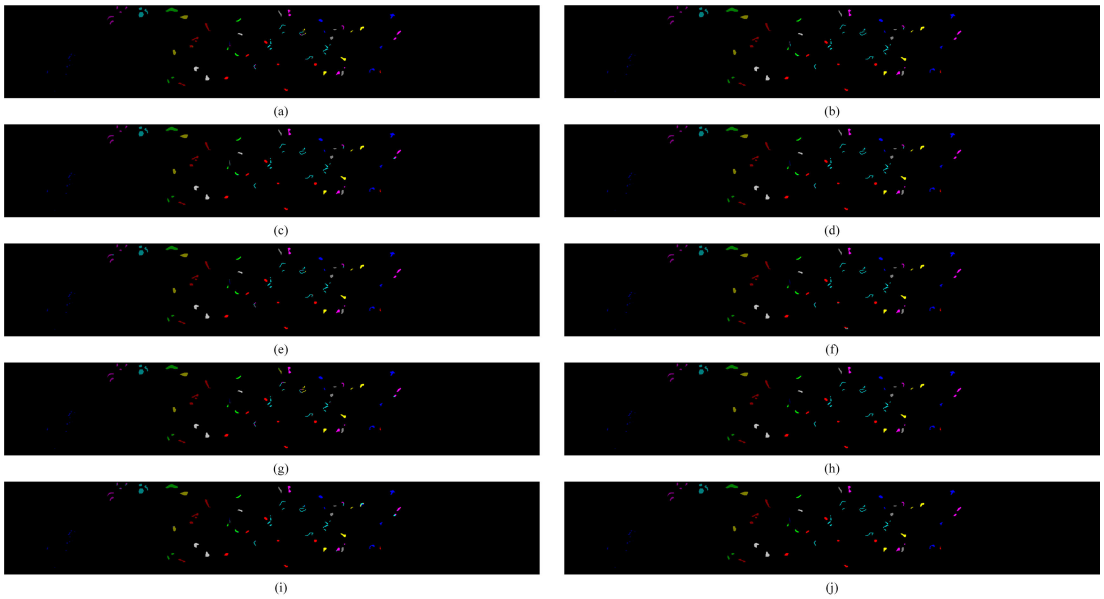


Fig. 13. Classification results for the BS dataset. (a) CDCNN (OA = 97.50%). (b) SSRN (OA = 99.23%). (c) FDSSC (OA = 99.37%). (d) DBMA (OA = 99.29%). (e) DBDA (OA = 99.48%). (f) SF (OA = 99.38%). (g) A-SPN (OA = 99.45%). (h) SSFTT (OA = 99.38%). (i) SS-MTr (OA = 99.32%). (j) MMAN (OA = 99.70%).

TABLE VII
ACCURACY COMPARISON OF DIFFERENT METHODS ON THE UP DATASET

Class No.	Accuracy of classification obtained by different comparison methods (in %)									
	CDCNN	SSRN	FDSSC	DBMA	DBDA	SF	A-SPN	SSFTT	SS-MTr	MMAN
1	95.28	98.18	99.80	98.80	99.23	94.76	98.80	98.72	99.70	**99.86**
2	98.99	99.33	99.96	99.65	99.88	98.38	99.92	99.92	99.95	**100.00**
3	93.14	96.04	**100.00**	92.26	99.65	83.28	95.21	98.04	97.04	97.79
4	98.73	**99.97**	99.89	98.59	99.05	97.00	98.45	96.96	97.20	98.26
5	**100.00**	**100.00**	99.92	99.71	99.92	99.84	99.69	98.16	97.77	99.92
6	96.45	98.95	99.98	99.28	99.54	98.85	99.91	99.02	99.87	**99.96**
7	96.27	99.83	99.95	99.82	99.97	93.96	**100.00**	99.40	99.00	99.88
8	92.03	97.62	93.92	96.42	95.27	96.61	98.04	98.13	98.50	**99.14**
9	99.36	**100.00**	99.77	99.59	99.78	99.22	98.61	95.86	87.57	98.56
OA(%)	97.15	98.89	99.37	98.73	99.22	99.35	99.23	99.01	99.06	**99.63**
AA(%)	96.69	98.88	99.25	98.26	99.33	96.76	98.75	98.25	97.40	**99.26**
Kappa	96.22	98.53	99.17	98.26	98.96	99.15	98.97	98.69	98.76	**99.51**

11 ground objects, the classification accuracy is 100%. The classification performance of Floodplain grasses 2 and Mixed mopane is significantly better than that of SF, DBDA, SS-MTr, and other methods, and the maximum OA difference between SF and SF is 19.09%, as shown in Fig. 13. However, the performance of all methods in the classification effect on Reeds1 is not good enough. The possible reason is that the distribution of this ground object is irregular and scattered, which makes it difficult for the model to learn and extract effective features. In short, the proposed method shows strong classification performance both in terms of overall performance and the classification accuracy of each category.

2) Results on the UP Dataset: On the UP dataset, the proposed method shows strong classification performance. Due to the powerful global feature extraction ability of transformer, the classification performance of all transformer-based comparison methods is strong, especially FDSSC, DBDA, and other methods, and the overall image classification rate is above 99%. As shown in Table VII, the classification performance of the MMAN method on Asphalt, Meadows, Bricks, and other four categories is better than other methods, which fully embodies the powerful ability of the proposed multiscale convolution and multiangle transformer to extract local and global features of HSI and the specific classification situation is shown in Fig. 14. In addition, compared with SSFTT, CDCNN, and SSRN perform poorly on the dataset, and the reason may be that CDCNN focuses too much on the spatial features of the image and does not effectively extract the spectral features of the image.

3) Results on the HU Dataset: On the HU dataset, the classification performance of the proposed method is generally higher than that of other compared methods, and the specific classification results are shown in Table VIII. Because MMAN exploits the excellent local feature extraction ability of the convolutional network and the modeling ability of the transformer network to model the relationship between global features, as well as the combination of multiscale input, its

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.

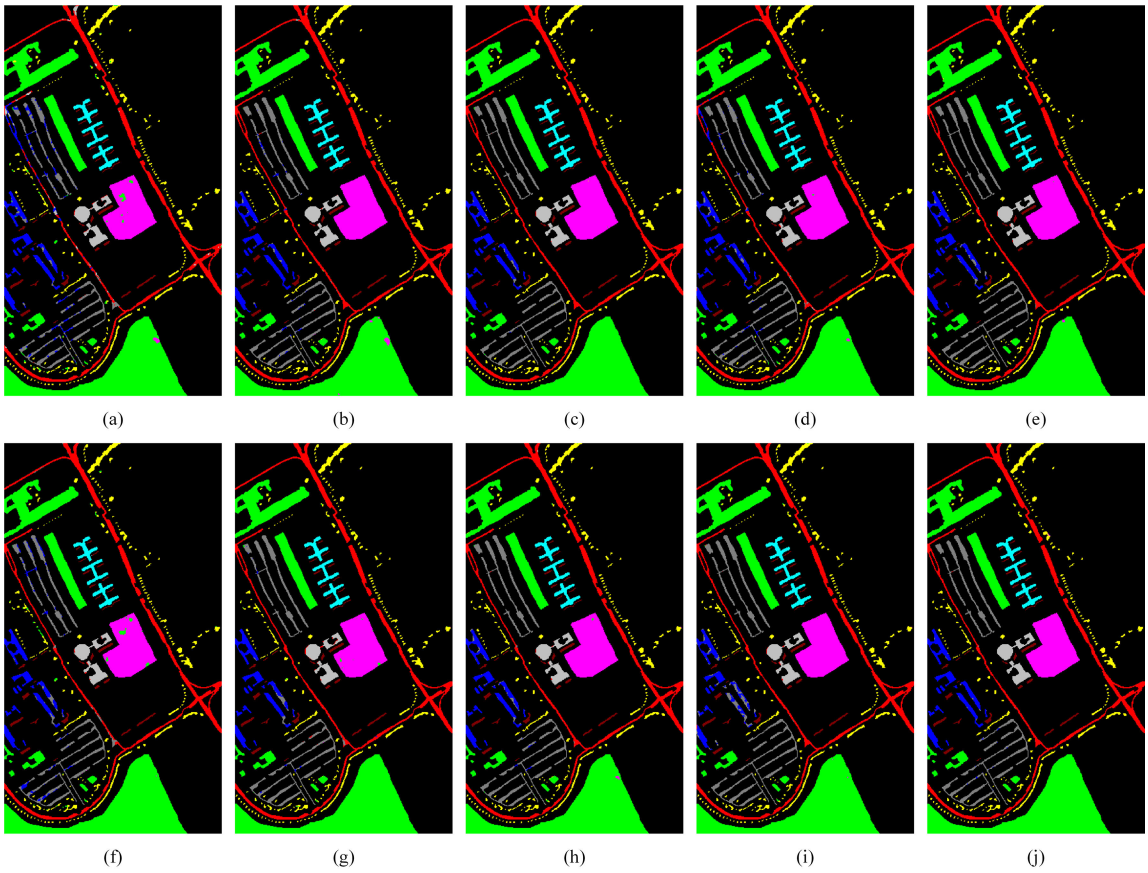


Fig. 14. Classification results for the UP dataset. (a) CDCNN (OA = 97.15%). (b) SSRN (OA = 98.89%). (c) FDSSC (OA = 99.37%). (d) DBMA (OA = 98.73%). (e) DBDA (OA = 99.22%). (f) SF (OA = 99.35%). (g) A-SPN (OA = 99.23%). (h) SSFTT (OA = 99.01%). (i) SS-MTr (OA = 99.06%). (j) MMAN (OA = 99.63%).

TABLE VIII
ACCURACY COMPARISON OF DIFFERENT METHODS ON THE HU DATASET

Class No.	Accuracy of classification obtained by different comparison methods (in %)									
	CDCNN	SSRN	FDSSC	DBMA	DBDA	SF	A-SPN	SSFTT	SS-MTr	MMAN
1	89.26	98.80	99.24	96.64	98.29	99.73	99.52	98.50	98.40	**99.87**
2	99.84	99.74	99.26	98.96	99.85	99.65	**99.96**	99.70	97.70	99.07
3	95.70	**100.00**	**100.00**	99.72	99.47	99.68	99.84	99.78	99.60	99.44
4	99.10	**99.56**	99.34	99.29	98.66	98.30	99.52	98.68	97.86	98.97
5	98.92	99.63	99.61	99.77	99.70	99.64	**100.00**	99.98	99.95	99.87
6	97.84	**100.00**	99.89	99.33	**100.00**	83.90	99.89	99.45	88.40	**100.00**
7	89.09	98.48	99.18	97.43	98.51	91.50	97.40	96.00	96.41	**99.47**
8	94.91	98.60	97.46	98.52	**99.74**	94.29	95.64	97.43	95.31	98.26
9	94.41	97.75	98.50	97.46	98.36	90.33	98.03	98.81	96.32	**99.11**
10	87.53	97.03	98.67	97.87	98.76	98.37	98.88	99.67	98.19	**99.82**
11	82.52	98.62	98.92	98.40	97.62	96.13	99.42	**99.57**	98.65	99.37
12	90.24	98.03	98.95	96.94	95.29	88.47	98.44	**99.31**	95.90	98.83
13	94.17	**98.85**	98.45	96.31	98.27	60.90	97.90	97.48	92.79	98.09
14	97.27	99.97	99.38	99.07	**100.00**	99.48	**100.00**	**100.00**	**100.00**	**100.00**
15	96.64	99.31	99.21	98.75	99.26	99.83	**100.00**	99.26	98.91	**100.00**
OA(%)	92.97	98.77	98.99	98.19	98.62	97.74	98.86	98.84	97.36	**99.30**
AA(%)	93.83	98.96	99.10	98.30	98.79	97.91	98.99	98.91	96.96	**99.34**
Kappa	92.40	98.67	98.91	98.04	98.79	97.55	98.77	98.74	97.15	**99.25**

performance has been greatly improved compared with other methods based on convolutional network and transformer. The specific classification diagram is shown in Fig. 15. Among them, the classification accuracy of healthy grass, water, residential and the other seven categories reached the best, and the maximum classification accuracy gap was 16.85%. This is because the HU dataset contains large and small ground object distributions, and the proposed method is only able to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.

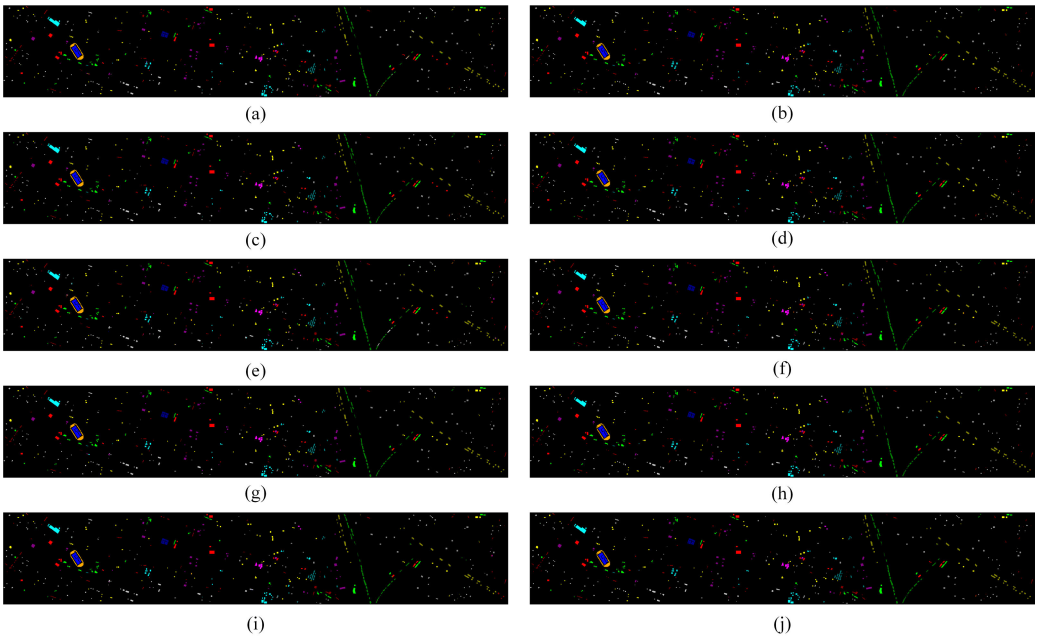


Fig. 15. Classification results of ground truth and comparison methods for the HU dataset. (a) CDCNN(OA = 92.97%). (b) SSRN(OA = 98.77%). (c) FDSSC(OA = 98.99%). (d) DBMA(OA = 98.19%). (e) DBDA(OA = 98.62%). (f) SF(OA = 97.74%). (g) A-SPN(OA = 98.86%). (h) SSFTT(OA = 98.84%). (i) SS-MTr(OA = 97.36%). (j) MMAN(OA=99.30%).

TABLE IX
ACCURACY COMPARISON OF DIFFERENT METHODS ON THE XZ DATASET

Class No.	Accuracy of classification obtained by different comparison methods (in %)									
	CDCNN	SSRN	FDSSC	DBMA	DBDA	SF	A-SPN	SSFTT	SS-MTr	MMAN
1	99.86	**99.85**	99.77	99.68	99.77	**99.85**	99.49	99.62	99.63	99.77
2	**100.00**	**100.00**	**100.00**	99.91	**100.00**	99.59	99.93	**100.00**	99.84	99.77
3	99.98	99.99	99.98	**100.00**	99.99	99.84	99.52	99.00	99.81	**100.00**
4	**100.00**	99.93	99.95	**100.00**	99.88	99.79	99.91	99.86	99.67	99.96
5	99.76	**99.93**	99.73	99.77	99.88	99.43	99.81	99.46	99.56	99.67
6	99.23	99.25	99.32	99.60	98.04	98.44	99.68	99.56	98.22	**100.00**
7	99.47	99.74	99.65	99.95	99.72	99.84	99.90	99.91	99.49	**99.95**
8	99.47	99.84	99.91	99.90	99.88	99.88	99.59	**99.97**	99.91	99.81
9	99.65	99.77	99.87	99.80	99.59	99.85	**99.93**	99.66	99.09	**99.93**
OA(%)	99.78	99.80	99.79	98.79	99.70	99.71	99.69	99.66	99.57	**99.81**
AA(%)	99.74	99.78	99.80	99.84	99.61	99.62	99.75	99.67	99.47	**99.87**
Kappa	99.72	99.75	99.73	99.74	99.62	99.63	99.60	99.57	99.45	**99.76**

effectively extract local and global features for different-sized ground objects.

4) Results on the XZ Dataset: Compared with the other three datasets, the XZ dataset has a higher spatial resolution of 0.73 m/pixel, which is more conducive to extracting image features. Therefore, the overall classification of most methods is relatively high, and the specific classification is shown in Table IX. However, the classification performance of DBDA is the lowest, only 98.79%. A possible explanation is that the effect of feature extraction of spatial dense blocks and spectral dense blocks is not sufficient, resulting in poor performance in high-resolution images. Since this dataset is mainly comprised of medium and large ground objects and has a high resolution, the classification performance of all methods is basically consistent under a large training proportion, as shown in Fig 16. The best classification performance is achieved by the proposed method, which is highly competitive with other representative methods.

E. Classification Performance Under Different Proportions of Training Samples

In this article, different proportions of training samples are selected to test the classification performance and robustness of the proposed method and the comparison method. Specifically, 1%, 2%, 3%, 4%, and 5% of the training sample proportion are selected for the UP dataset, and 2%, 4%, 6%, 8%, and 10% of the training sample proportion are selected for the BS, HU, and XZ datasets. The classification results for different training ratios are shown in Fig. 17. As can be seen from the line chart, in Fig 17(a), DBDA performs best when the ratio is 2% and 4%. As the proportion of training samples increases, the classification performance of the pro-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.

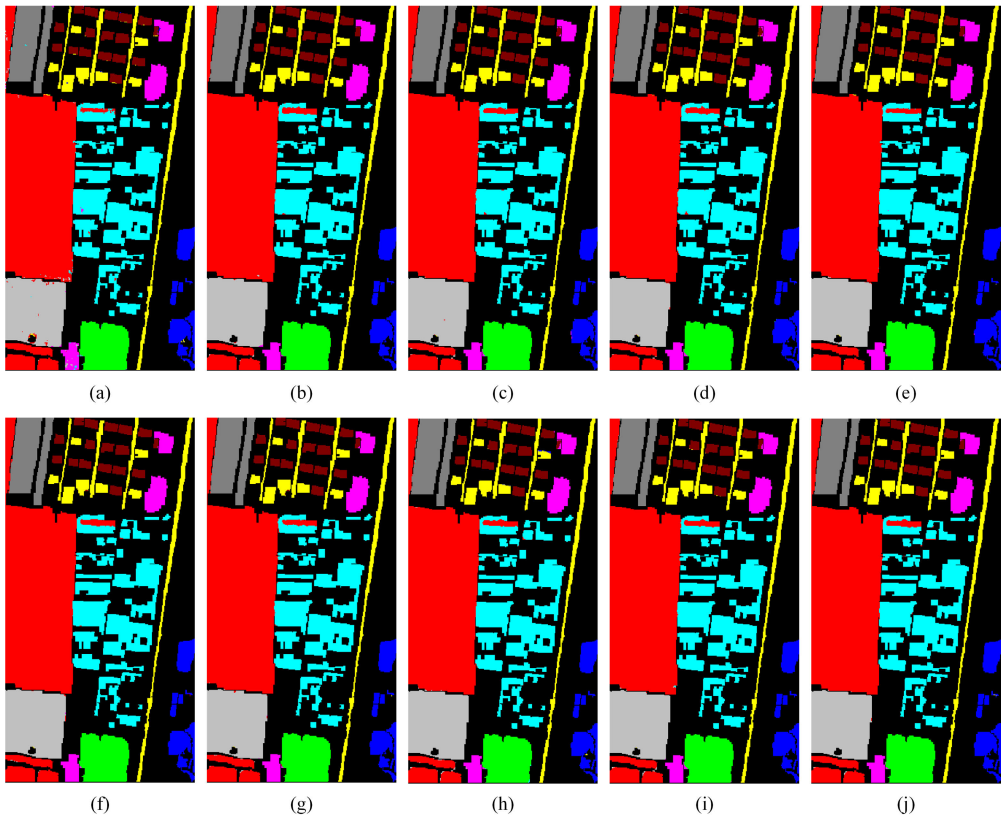


Fig. 16. Classification results for the XZ dataset. (a) CDCNN(OA = 99.78%). (b) SSRN(OA = 99.80%). (c) FDSSC(OA = 99.79%). (d) DBMA(OA = 99.79%). (e) DBDA(OA = 99.70%). (f) SF(OA = 99.71%). (g) A-SPN(OA = 99.69%). (h) SSFTT(OA = 99.66%). (i) SS-MTr(OA = 99.57%). (j) MMAN(OA = 99.81%).

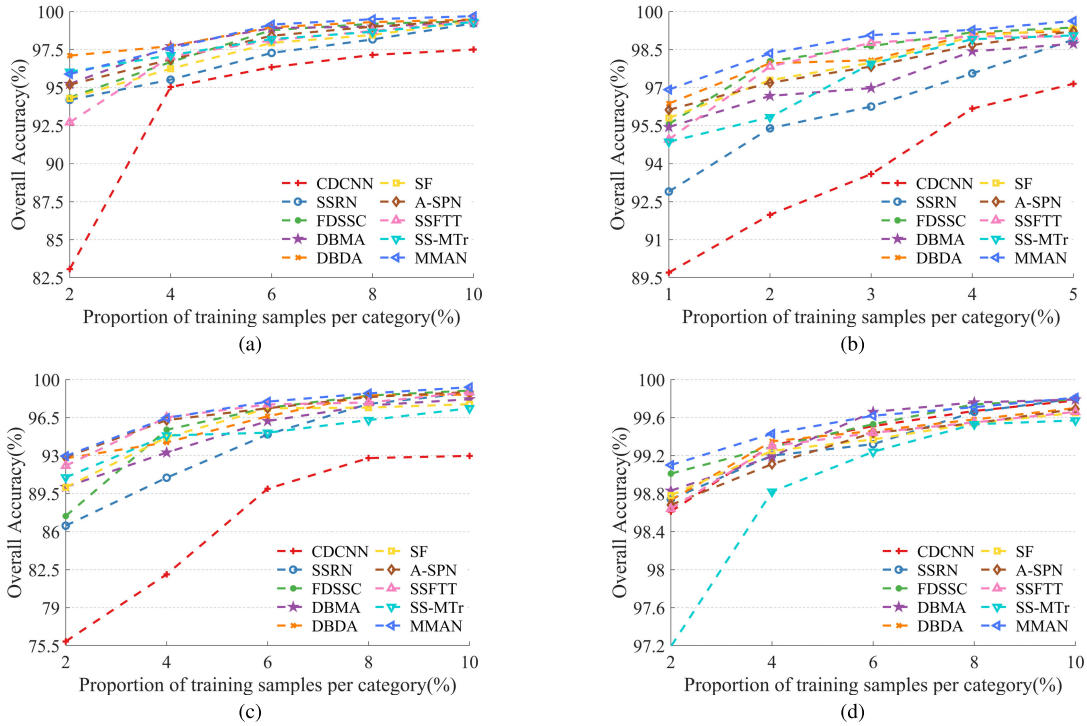


Fig. 17. Performance of our proposed method and other comparison methods on different datasets with different training samples. (a) BS dataset. (b) UP dataset. (c) HU dataset. (d) XZ dataset.

posed method quickly surpasses DBDA and maintains a steady increase. In Fig. 17(b)–(d), the classification performance of the proposed method is always in the leading position despite the strong competitiveness of the comparison methods SSFTT, DBDA, and other methods. According to the spatial resolution of the dataset, it can be seen that the BS dataset has the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.

TABLE X
RUNNING TIME OF DIFFERENT METHODS ON DIFFERENT DATASETS

Datasets	Times	computation time of the compared methods (in seconds)									
		CDCNN	SSRN	FDSSC	DBMA	DBDA	SF	A-SPN	SSFTT	SS-MTr	MMAN
BS	Training	13.86	48.88	64.26	56.28	152.34	138.45	**6.45**	23.41	98.65	35.57
	Test	**0.60**	0.97	1.54	1.25	4.54	1.71	1.11	0.65	1.98	1.15
UP	Training	92.26	179.15	255.85	114.98	83.35	925.34	**57.17**	163.93	499.45	233.95
	Test	**7.41**	11.02	18.16	13.72	19.63	11.09	12.48	11.06	13.48	15.91
HU	Training	129.79	74.45	152.45	224.04	215.86	459.16	39.33	**26.18**	564.36	93.56
	Test	2.96	3.18	6.50	5.60	7.12	2.66	1.46	**0.71**	5.53	3.13
XZ	Training	942.30	307.59	3327.13	3869.21	99.62	6331.73	**97.40**	98.02	6621.31	435.02
	Test	29.85	27.58	85.19	86.02	95.44	124.25	5.29	**2.44**	123.60	14.52

TABLE XI
TIME COMPLEXITY ANALYSIS

	Times(s)	SA	CSA
BS	Training	85.88	**35.57**
	Test	2.84	**1.15**
UP	Training	672.76	**233.95**
	Test	48.25	**15.91**
HU	Training	341.8	**93.56**
	Test	14.94	**3.13**
XZ	Training	1019.93	**435.02**
	Test	34.72	**14.52**

TABLE XII
OA AND STANDARD DEVIATION OF THE DIFFERENT COMPONENTS OF THE PROPOSED METHOD

MLF		✓			✓	✓	
MGF			✓			✓	
ViT				✓	✓		
MMAN							✓
Datasets	BS	93.54	92.47	91.56	98.17	99.34	**99.70**
	UP	94.15	92.36	90.78	97.42	99.23	**99.63**
	HU	91.96	90.79	89.13	97.26	98.89	**99.30**
	XZ	97.63	97.11	95.25	98.03	99.68	**99.81**

lowest spatial resolution, so in the case of a low training ratio, the proposed method is slightly lower than DBDA and other methods. In the case of high spatial resolution XZ, HU, and UP datasets, the OAs of MMAN have always been leading. This experiment fully demonstrates that the proposed method has stronger robustness and advantages in HSI classification.

F. Analysis of Computation Time

To analyze the computational complexity of the proposed method, we analyze the computational cost of all methods on the four datasets BS, UP, HU, and XZ, and the sample calculation ratio adopted is 10%, and the computational cost results are shown in Table X. It can be seen from the data in the table that the training time of MMAN is competitive among all methods, only the SSFTT and A-SPN methods have better training time than the proposed method, and the other methods have longer or comparable training time than the proposed method. This is because SSFTT uses a lightweight feature extraction network, and the network architecture of A-SPN is also very simple, which can effectively improve the convergence level. The multiscale feature extraction is equivalent to multiple-branch extraction and increases the training time to some extent. In terms of test time, the proposed method also has considerable advantages. Except for SSFTT, CDCNN, and A-SPN, which are slightly better than MMAN, the test time performance of other methods is not better than that of the proposed method. In general, the time complexity of the proposed method is small, which is fully proved by the performance on four datasets, and the network structure to reduce the time complexity will continue to be explored in future work.

Moreover, in the case of the same other structure, we compare the time complexity of the original self-attention mechanism and the time complexity of the correlation self-attention proposed in this article on four datasets. Table XI shows that the training time and testing time of correlation self-attention proposed in this article on the four datasets are shorter than that of original attention, and the maximum difference of training time is 3.65 times, and the maximum difference of testing time is 4.77 times. The above results fully show that the computational complexity of the proposed correlation self-attention is drastically reduced compared to the original self-attention.

G. Ablation Experiment

In this section, to fully verify the effectiveness of MMAN, we will fully verify the effectiveness of each part for HSI classification from the perspective of MLF, MGF, and multiscale structure involved in this method and compare with the ViT without improvement, to prove the superior performance of the structure proposed by this method. We conduct experimental verification on four datasets, the proportion of training samples in the verification experiment is 10%, the evaluation index is OA, and the specific results of each part are shown in Table XII.

As can be seen from the table, when only the MLF part is used, its classification performance is fine. This is because the pixels in HSI usually have a strong spatial correlation with each other, which means that there is a clear relationship between the features of neighboring pixels, and the convolutional network performs well in handling spatial locality, so it can effectively capture local features such as texture, shape, and edge. When only the MGF part is used, its overall classification performance is slightly worse than that obtained using only MLF. A possible reason is that MGF pays more attention to the global context information and ignores the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.

local detail features. When only ViT is used, its classification performance is not comparable to MGF, because MGF takes into account the specific distribution of features, and the largest OA gap between the two reaches 1.86%. Different patch sizes were set depending on the spatial resolution of the dataset. For the BS and UP datasets, we set patches of size $(2\tau+1)\times(2\tau+1)$, $(2\tau+3)\times(2\tau+3)$, $(2\tau+5)\times(2\tau+5)$, where $\tau=4$. For the HU and XZ datasets, we set patches of size $(2\tau+1)\times(2\tau+1)$, $(2\tau+3)\times(2\tau+3)$, $(2\tau+5)\times(2\tau+5)$, where τ is equal to 5. When MLF and ViT are combined, the overall classification performance is significantly improved, which is because the combined model can comprehensively focus on spatial local features and spectral global features, which is more conducive to the classification of HSIs. When MLF and MGF are combined, the performance is obviously much better. When using the method proposed in this article, the combination of MLF, MGF, and multiscale is realized, which not only considers the local features and global features, but also fully mines the relationship between features from the perspective of feature distribution and distribution size. From the performance of the four datasets, it can be seen that the proposed method comprehensively outperforms the previous module combinations and further improves the classification performance.

IV. Conclusion

In this article, we propose a novel and effective HSI classification method based on the combination of the transformer and convolutional networks, which is mainly composed of three parts: hybrid convolutional network, multiangle self-attention network, and MFF. The hybrid convolutional network mainly extracts and learns the local spatial features in the HSI. Then, we introduce an improved transformer mechanism to capture the spectral correlation of HSI data, to make full use of spectral and spatial information, thus improving the classification performance, reducing the number of parameters, and improving the model's interpretability. Finally, by fusing multiscale features, the model can comprehensively use the feature details and the overall structure and contextual information to understand the image content more comprehensively and improve the classification performance, as well as the robustness and interpretability.

In future research, we plan to explore the adaptability of feature information fusion methods and further reduce the computational complexity of the attention mechanism, which can be explored from the sparsity of attention. Moreover, the input form of this article is regular patches, while most ground objects are distributed in irregular shapes. Regular shape input will tear the correlation between features. In view of the form of input data, we consider superpixel segmentation as the input of the model to explore the detailed information and diversity of HSIs.

References

[1] J. A. Richards and X. Jia, "Using suitable neighbors to augment the training set in hyperspectral maximum likelihood classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 4, pp. 774–777, Oct. 2008.

[2] J. Peng, L. Li, and Y. Y. Tang, "Maximum likelihood estimation-based joint sparse representation for the classification of hyperspectral remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 6, pp. 1790–1802, Jun. 2019.

[3] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[4] Q. Ye, P. Huang, Z. Zhang, Y. Zheng, L. Fu, and W. Yang, "Multiview learning with robust double-sided twin SVM," *IEEE Trans. Cybern.*, vol. 52, no. 12, pp. 12745–12758, Dec. 2022.

[5] Y.-N. Chen, T. Thaipisutikul, C.-C. Han, T.-J. Liu, and K.-C. Fan, "Feature line embedding based on support vector machine for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 1, p. 130, Jan. 2021.

[6] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar. 2005.

[7] M. Sheykhmousa, M. Mahdianpari, H. Ghanbari, F. Mohammadimanesh, P. Ghamisi, and S. Homayouni, "Support vector machine versus random forest for remote sensing image classification: A meta-analysis and systematic review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 6308–6325, 2020.

[8] S. Prasad and L. M. Bruce, "Limitations of principal components analysis for hyperspectral target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 4, pp. 625–629, Oct. 2008.

[9] A. Villa, J. A. Benediktsson, J. Chanussot, and C. Jutten, "Hyperspectral image classification with independent component discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 4865–4876, Dec. 2011.

[10] T. V. Bandos, L. Bruzzone, and G. Camps-Valls, "Classification of hyperspectral images with regularized linear discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 3, pp. 862–873, Mar. 2009.

[11] W. Liao, A. Pizurica, P. Scheunders, W. Philips, and Y. Pi, "Semisupervised local discriminant analysis for feature extraction in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 1, pp. 184–198, Jan. 2013.

[12] Q. Wang, Z. Meng, and X. Li, "Locality adaptive discriminant analysis for spectral–spatial classification of hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2077–2081, Nov. 2017.

[13] R. Hang, Q. Liu, D. Hong, and P. Ghamisi, "Cascaded recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5384–5394, Aug. 2019.

[14] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[15] Y. Zhan, D. Hu, Y. Wang, and X. Yu, "Semisupervised hyperspectral image classification based on generative adversarial networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 212–216, Feb. 2018.

[16] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[17] X. Liao, B. Tu, J. Li, and A. Plaza, "Class-wise graph embedding-based active learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522813, doi: 10.1109/TGRS.2023.3309032.

[18] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Feb. 2015, doi: 10.1155/2015/258619.

[19] W. Zhao and S. Du, "Spectral–spatial feature extraction for hyperspectral image classification: A dimension reduction and deep learning approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4544–4554, Aug. 2016.

[20] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[21] M. Kanthi, T. H. Sarma, and C. S. Bindu, "A 3D-deep CNN based feature extraction and hyperspectral image classification," in *Proc. IEEE India Geosci. Remote Sens. Symp. (InGARSS)*, Dec. 2020, pp. 229–232, doi: 10.1109/InGARSS48198.2020.9358920.

[22] J. Yang, Y.-Q. Zhao, and J. C. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.

[23] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.

[24] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2017.

[25] Z. Xue, Z. Liu, and M. Zhang, "DSR-GCN: Differentiated-scale restricted graph convolutional network for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504918, doi: 10.1109/TGRS.2023.3253248.

[26] R. Dian, A. Guo, and S. Li, "Zero-shot hyperspectral sharpening," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12650–12666, Oct. 2023.

[27] R. Dian, T. Shan, W. He, and H. Liu, "Spectral super-resolution via model-guided cross-fusion network," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–12, Jan. 2023, doi: 10.1109/TNNLS.2023.3238506.

[28] M. E. Paoletti, J. M. Haut, R. Fernandez-Beltran, J. Plaza, A. J. Plaza, and F. Pla, "Deep pyramidal residual networks for spectral–spatial hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 740–754, Feb. 2019.

[29] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, May 2020.

[30] C. Zhao, W. Zhu, and S. Feng, "Superpixel guided deformable convolution network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3838–3851, 2022.

[31] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inform. Process. Syst.*, 2017, pp. 5998–6008.

[32] C. Zhou, Q. Shi, D. He, B. Tu, H. Li, and A. Plaza, "Spectral–spatial sequence characteristics-based convolutional transformer for hyperspectral change detection," *CAAI Trans. Intell. Technol.*, vol. 8, no. 4, pp. 1237–1257, Dec. 2023, doi: 10.1049/cit2.12226.

[33] B. Tu, Q. Ren, J. Li, Z. Cao, Y. Chen, and A. Plaza, "NCGLF2: Network combining global and local features for fusion of multisource remote sensing data," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102192.

[34] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Sep. 2019.

[35] R. Child, S. Gray, A. Radford, and I. Sutskever, "Generating long sequences with sparse transformers," 2019, *arXiv:1904.10509*.

[36] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[37] Z. Xue, M. Zhang, Y. Liu, and P. Du, "Attention-based second-order pooling network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9600–9615, Nov. 2021.

[38] Z. Xue, Q. Xu, and M. Zhang, "Local transformer with spatial partition restore for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4307–4325, 2022.

[39] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615, doi: 10.1109/TGRS.2021.3130716.

[40] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[41] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.

[42] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214, doi: 10.1109/TGRS.2022.3144158.

[43] C. Zhao et al., "Hyperspectral image classification with multi-attention transformer and adaptive superpixel segmentation-based active learning," *IEEE Trans. Image Process.*, vol. 32, pp. 3606–3621, 2023.

[44] B. Tu, X. Liao, Q. Li, Y. Peng, and A. Plaza, "Local semantic feature aggregation-based transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536115, doi: 10.1109/TGRS.2022.3201145.

[45] Y. Li et al., "Efficient and explicit modelling of image hierarchies for image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 18278–18289.

[46] X. Huang and L. Zhang, "A comparative study of spatial approaches for urban mapping using hyperspectral ROSIS images over Pavia City, northern Italy," *Int. J. Remote Sens.*, vol. 30, no. 12, pp. 3205–3221, Jun. 2009.

[47] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-Hi: UAV-borne hyperspectral with high spatial resolution (H^2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.

[48] Y. Zhong et al., "Mini-UAV-borne hyperspectral remote sensing: From observation and processing to applications," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 4, pp. 46–62, Dec. 2018.

[49] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.

[50] W. Wang, S. Dou, Z. Jiang, and L. Sun, "A fast dense spectral–spatial convolution network framework for hyperspectral images classification," *Remote Sens.*, vol. 10, no. 7, p. 1068, Jul. 2018.

[51] W. Ma, Q. Yang, Y. Wu, W. Zhao, and X. Zhang, "Double-branch multi-attention mechanism network for hyperspectral image classification," *Remote Sens.*, vol. 11, no. 11, p. 1307, Jun. 2019.

[52] R. Li, S. Zheng, C. Duan, Y. Yang, and X. Wang, "Classification of hyperspectral image based on double-branch dual-attention mechanism network," *Remote Sens.*, vol. 12, no. 3, p. 582, Feb. 2020.

[53] L. Huang, Y. Chen, and X. He, "Spectral–spatial masked transformer with supervised and contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508718.

[54] K. Ding, T. Lu, W. Fu, S. Li, and F. Ma, "Global–local transformer network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541213, doi: 10.1109/TGRS.2022.3216319.



Jianghong Hu (Graduate Student Member, IEEE) received the B.S. degree in electrical engineering and automation from Hunan Institute of Science and Technology, Yueyang, China, in 2020, where he is currently pursuing the M.S. degree in information and communication engineering.

His research interests include deep learning and hyperspectral image processing.



Bing Tu (Senior Member, IEEE) received the M.S. degree in control science and engineering from Guilin University of Technology, Guilin, China, in 2009, and the Ph.D. degree in mechatronic engineering from Beijing University of Technology, Beijing, China, in 2013.

From 2015 to 2016, he was a Visiting Researcher with the Department of Computer Science and Engineering, University of Nevada, Reno, NV, USA, which is supported by the China Scholarship Council. He is currently a Full Professor with Nanjing University of Information Science and Technology, Nanjing, China. His research interests include sparse representation, pattern recognition, and analysis in remote sensing.

Dr. Tu is an Associate Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.



Qi Ren (Student Member, IEEE) received the B.S. degree in computer science and technology from Shanxi Datong University, Datong, China, in 2020, and the M.S. degree in information and communication engineering from Hunan Institute of Technology, Yueyang, China, in 2023. He is currently pursuing the Ph.D. degree with the School of Computer and Software, Shenzhen University, Shenzhen, China.

His research interests include hyperspectral image processing, pattern recognition, and spectral construction.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.



Xiaolong Liao (Graduate Student Member, IEEE) received the B.S. degree in automation from Hunan Institute of Science and Technology, Yueyang, China, in 2021, where he is currently pursuing the M.S. degree in information and communication engineering.

His research interests include hyperspectral image processing and remote-sensing image registration.



Zhaolou Cao (Member, IEEE) received the B.S. degree in mechanical engineering and the Ph.D. degree in measurement technology and instruments from the University of Science and Technology of China, Hefei, China, in 2009 and 2014, respectively.

He is currently an Associate Professor with Nanjing University of Information Science and Technology, Nanjing, China. He has authored or coauthored more than 30 publications (articles and inventions). His research interests include electromagnetic theory, optical metrology, and imaging techniques.



Antonio Plaza (Fellow, IEEE) received the M.Sc. and Ph.D. degrees in computer engineering from the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, University of Extremadura, Cáceres, Spain, in 1999 and 2002, respectively.

He is currently the Head of the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, University of Extremadura. He has authored more than 600 publications, including more than 200 JCR journal articles (more than 160 in IEEE journals), 23 book chapters, and around 300 peer-reviewed conference proceedings papers. His research interests include hyperspectral data processing and parallel computing of remote sensing data.

Dr. Plaza was a member of the Editorial Board of IEEE GEOSCIENCE AND REMOTE SENSING NEWSLETTER from 2011 to 2012 and *IEEE Geoscience and Remote Sensing Magazine* in 2013. He was also a member of the Steering Committee of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS). He is also a fellow of IEEE for his contributions to hyperspectral data processing and parallel computing of Earth observation data. He was a recipient of the recognition of Best Reviewers of IEEE GEOSCIENCE AND REMOTE SENSING LETTERS in 2009 and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING in 2010, for which he has served as an Associate Editor from 2007 to 2012. He was also a recipient of the Best Column Award of *IEEE Signal Processing Magazine* in 2015, the 2013 Best Paper Award of the JSTARS journal, and the Most Highly Cited Paper (2005–2010) in *Journal of Parallel and Distributed Computing*. He received best paper awards at the IEEE International Conference on Space Technology and the IEEE Symposium on Signal Processing and Information Technology. He has served as the Director of Education Activities for the IEEE Geoscience and Remote Sensing Society (GRSS) from 2011 to 2012 and the President of the Spanish Chapter of IEEE GRSS from 2012 to 2016. He has reviewed more than 500 manuscripts for more than 50 different journals. He served as the Editor-in-Chief for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING from 2013 to 2017. He has guest-edited ten special issues on hyperspectral remote sensing for different journals. He is also an Associate Editor of IEEE ACCESS (receiving recognition as an Outstanding Associate Editor of the journal in 2017). Additional information is available at http://www.umbc.edu/rssipl/people/aplaza.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:54:07 UTC from IEEE Xplore. Restrictions apply.